DO NOW — RECALL

Lesson 146 · Network Threats: Attack Paths and Conseq

Name: _____ Class: _____ Date: _____

1. When packets arrive out of order, what can TCP do?

2. Which layer in TCP/IP handles IP routing?

3. What data structure follows FIFO?

4. What is the result of an AND gate with inputs 1 and 0?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Use sequence information to reassemble the byte stream correctly.

2. Model answer

The internet layer.

3. Model answer

A queue.

4. Model answer

0.

By the end, you can…

• Distinguish malware, phishing, interception and denial-of-service threats without conflating them.

• Trace how a network attack reaches an asset and identify the vulnerability involved.

• Explain an appropriate defence for the particular threat rather than naming generic 'security'.

Vocabulary — know these terms

phishing — Social engineering to obtain information or actions through deception.

malware — Software designed to perform harmful or unauthorised actions.

DDoS — Distributed denial-of-service intended to disrupt availability.

CIA triad — Confidentiality, integrity and availability.

NEW LEARNING

Read, annotate and explain

Entry point

Suspicious link, exposed service or weak account.

Threat action

Credential theft, malware execution or traffic flooding.

Effect on service

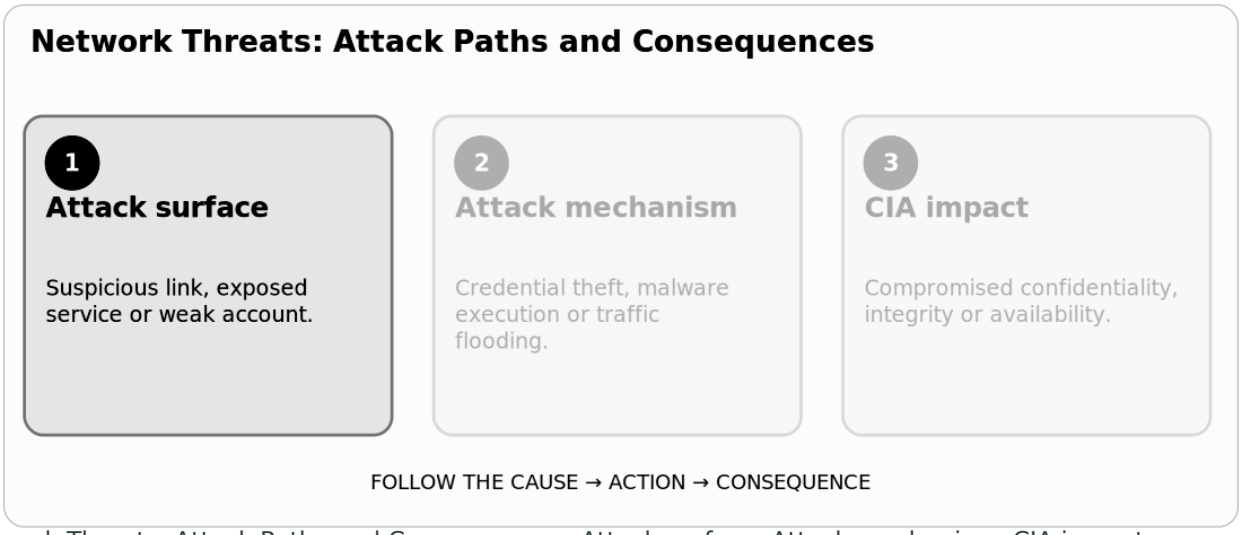
Compromised confidentiality, integrity or availability.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

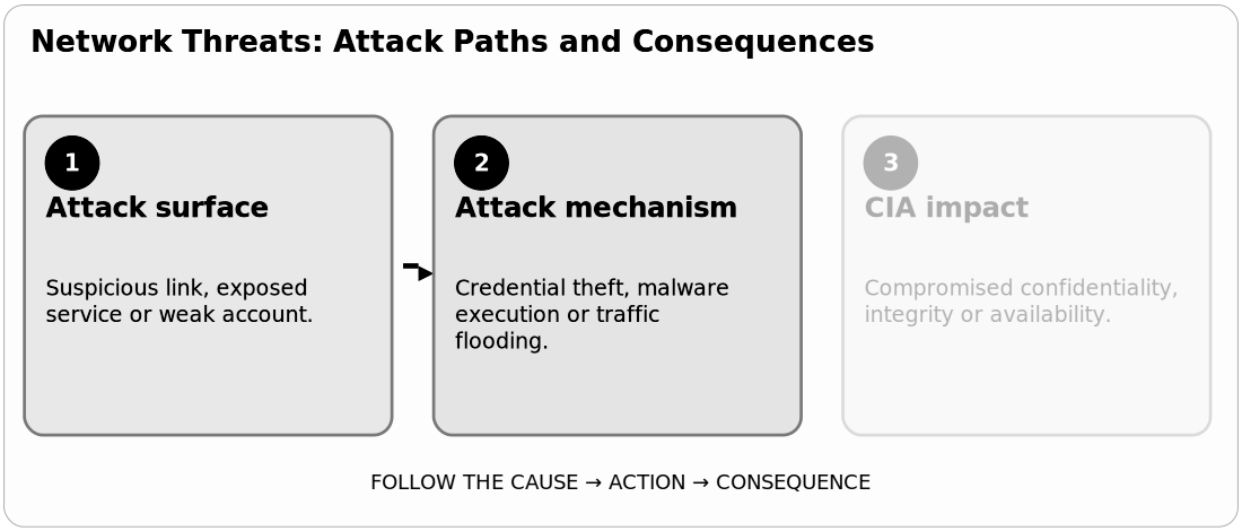
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Network Threats: Attack Paths and Consequences: Attack surface, Attack mechanism, CIA impact

New Learning

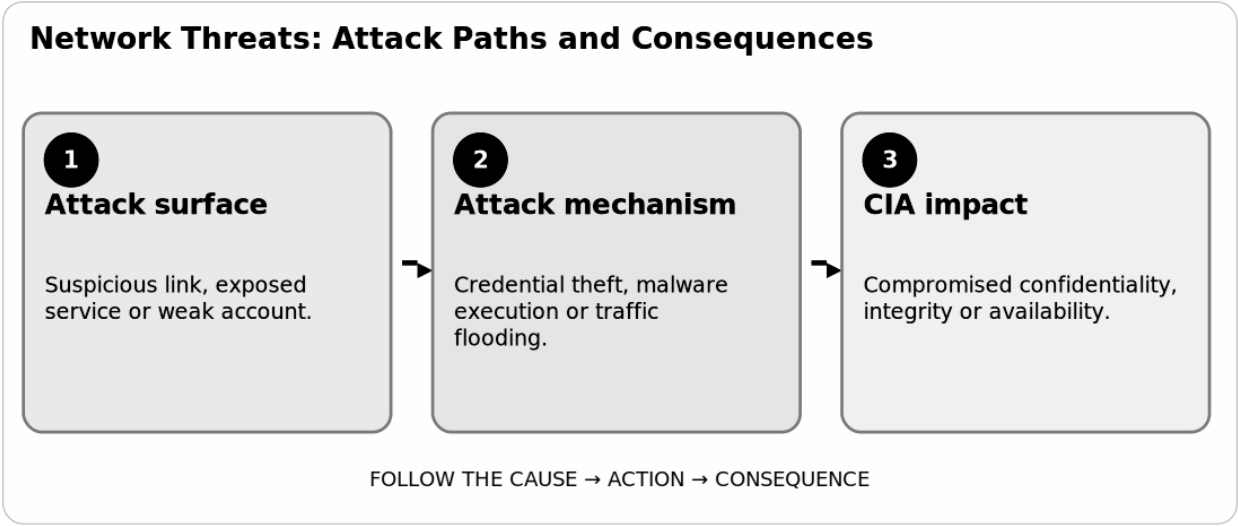


Network Threats: Attack Paths and Consequences: Attack surface, Attack mechanism, CIA impact

New Learning · original lesson diagrams

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New Learning



Network Threats: Attack Paths and Consequences: Attack surface, Attack mechanism, CIA impact

I DO — WORKED EXAMPLE

Follow the worked method

A pupil receives a fake school login link and submits a password. Explain the attack.

Fake login email

Identify the facts and make an initial prediction.

Credentials submitted

Explain the mechanism or trace the intermediate stage.

Account access at risk

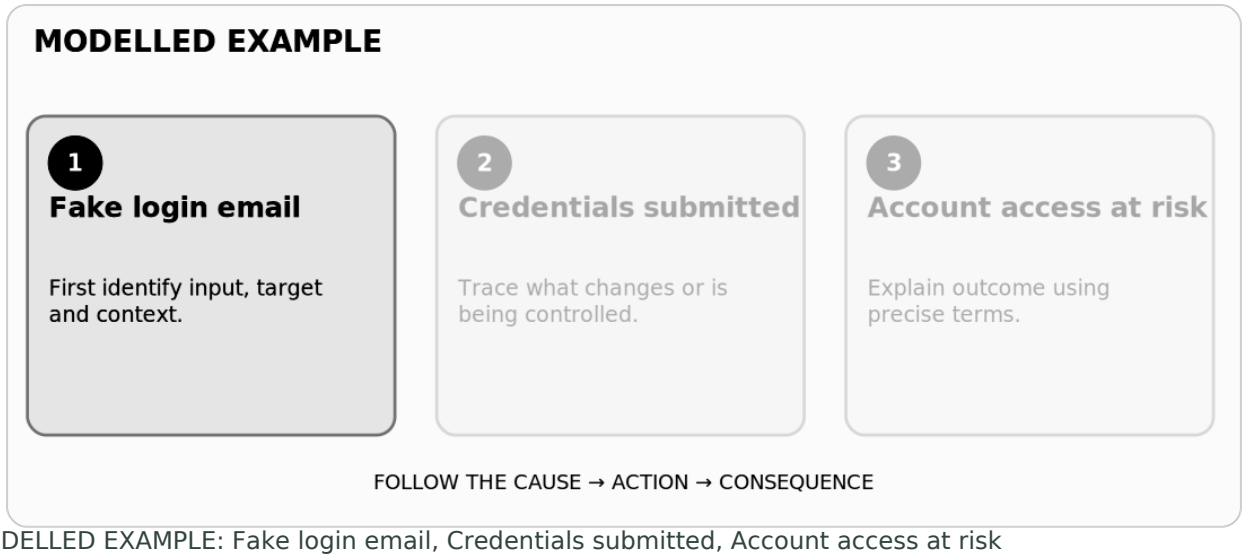
Phishing tricks the pupil into visiting an attacker-controlled form. The secret is disclosed, then the attacker may attempt unauthorised account access. Awareness and phishing-resistant MFA help, while a firewall alone does not prevent users entering secrets into fake sites.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil receives a fake school login link and submits a password. Explain the attack.

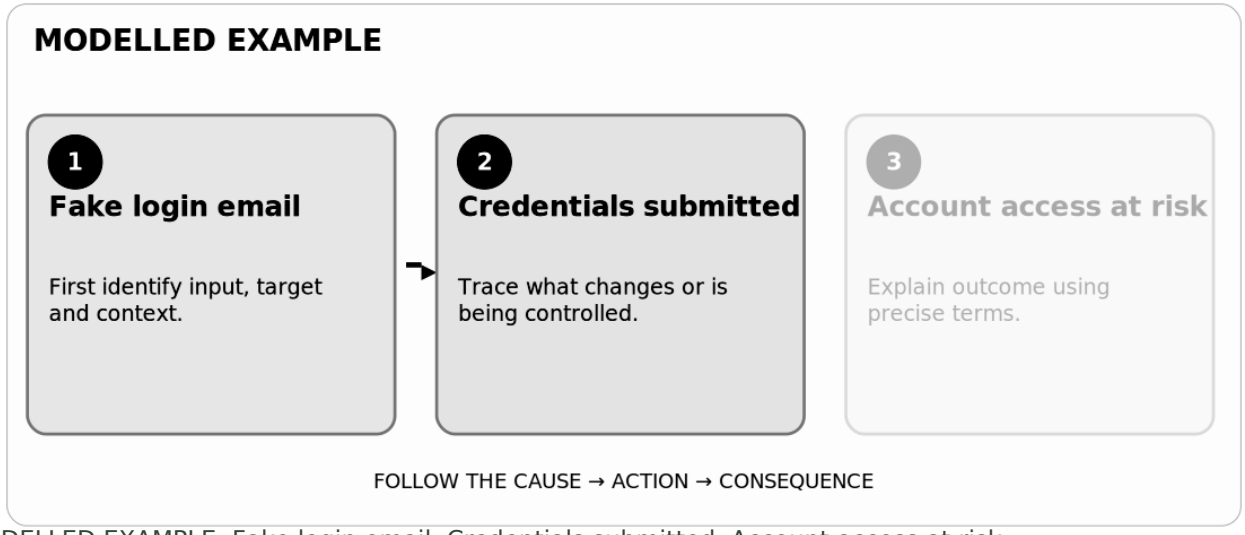
Identify the facts and make an initial prediction.



MODELLED EXAMPLE: Fake login email, Credentials submitted, Account access at risk

A pupil receives a fake school login link and submits a password. Explain the attack.

Identify the facts and make an initial prediction.



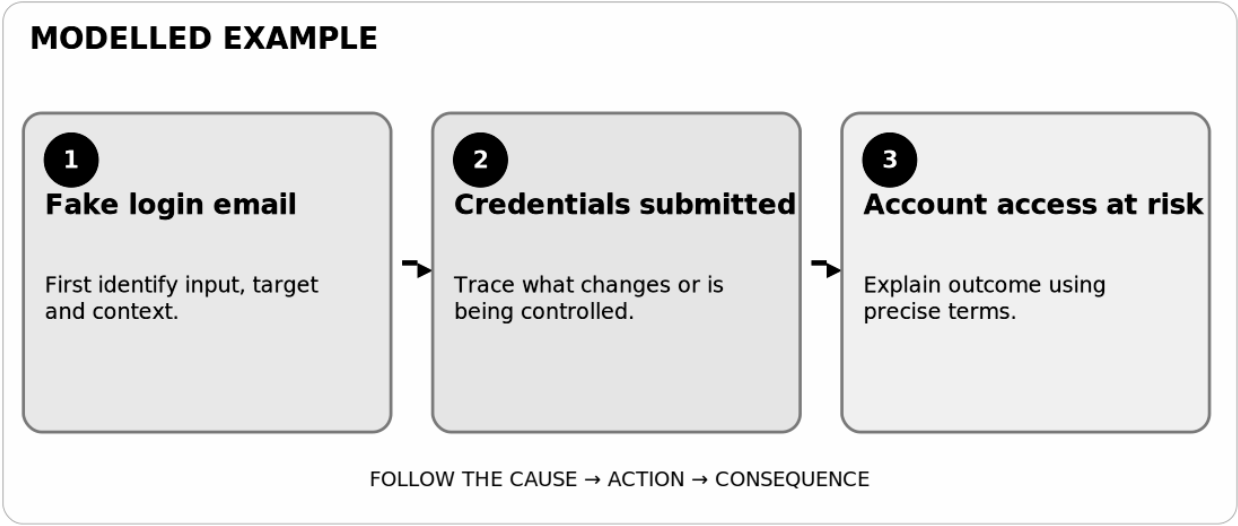
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I Do · original lesson diagrams

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WE DO — GUIDED PRACTICE

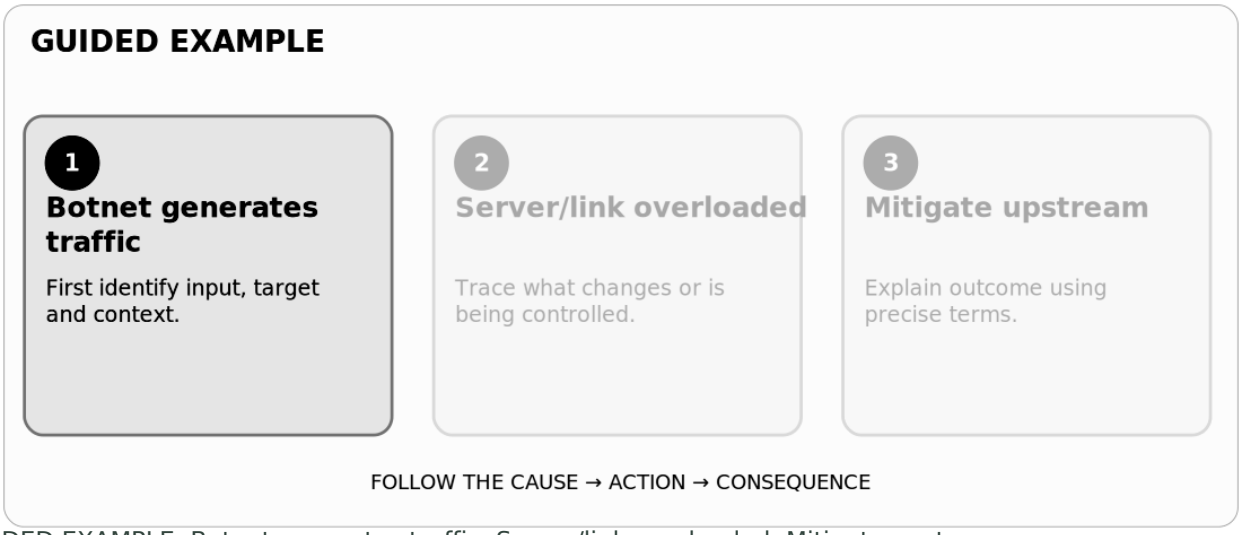
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A public learning site stops responding during a flood of requests from many infected machines. Name the threa

Identify the facts and make an initial prediction.



GUIDED EXAMPLE: Botnet generates traffic, Server/link overloaded, Mitigate upstream

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Botnet generates traffic Identify the facts and make an initial prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain how phishing differs from malware and give one example of each. [6]

Question 2

A school site is reachable but pupils' account details are exposed. Identify the affected security property and two controls. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A network suffers DDoS and suspected password theft simultaneously. Prioritise two distinct mitigations and justify why a single firewall rule is inadequate. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Encrypting a site with HTTPS is sufficient to prevent a DDoS attack.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

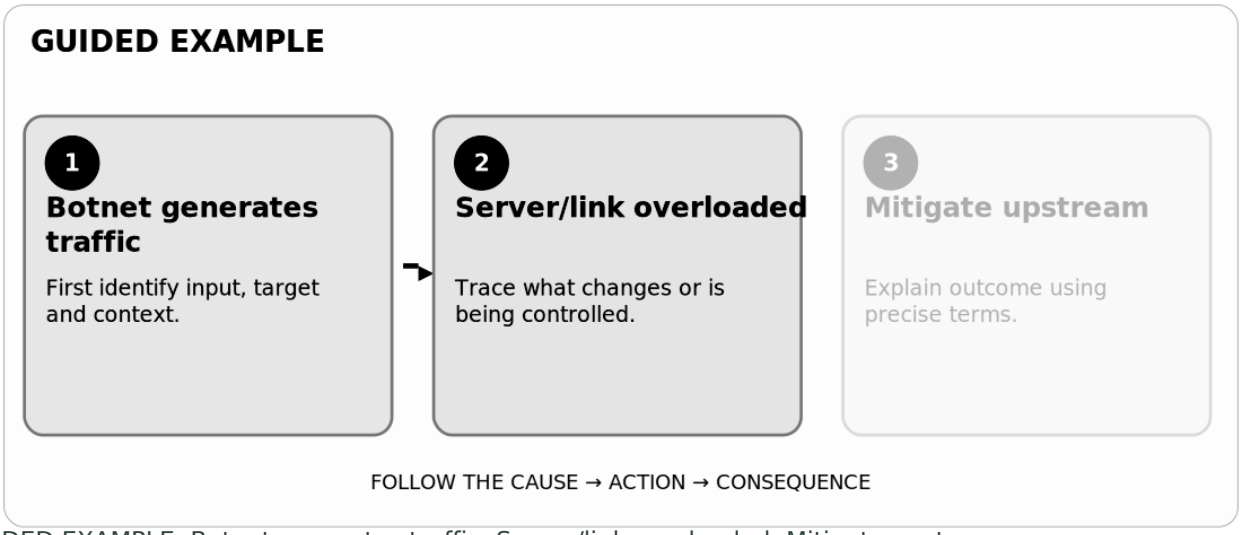
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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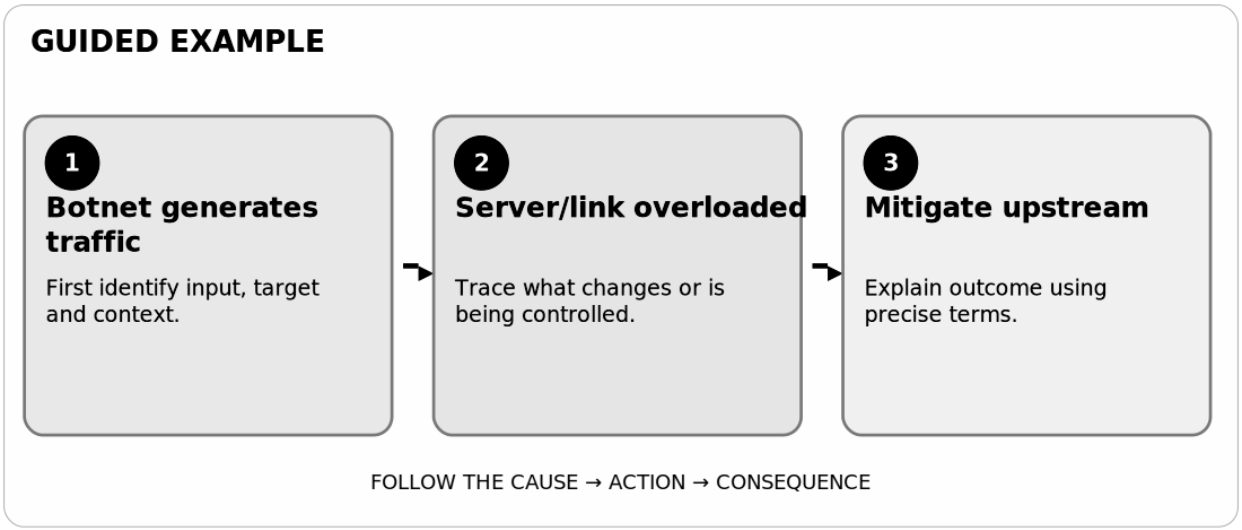
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GUIDED EXAMPLE: Botnet generates traffic, Server/link overloaded, Mitigate upstream

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GUIDED EXAMPLE: Botnet generates traffic, Server/link overloaded, Mitigate upstream

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Server/link overloaded

Explain the mechanism or trace the intermediate stage.

Mitigate upstream

A distributed denial-of-service (DDoS) attack aims to exhaust bandwidth or service resources. Upstream rate-limiting, DDoS protection and filtering may reduce impact; encrypting the website does not stop resource saturation.

IMPROVE: Explain one correction to your own reasoning.